SOKOINE UNIVERSITY OF AGRICULTURE
School of Engineering and Technology | Department of Agricultural Engineering

AEN 214: INTRODUCTION TO ELECTRONICS — LECTURE 6

Field Effect Transistor (FET)

JFETs, MOSFETs, Gate Charge Dynamics, Power MOSFETs & Solar Irrigation Pump Inverters

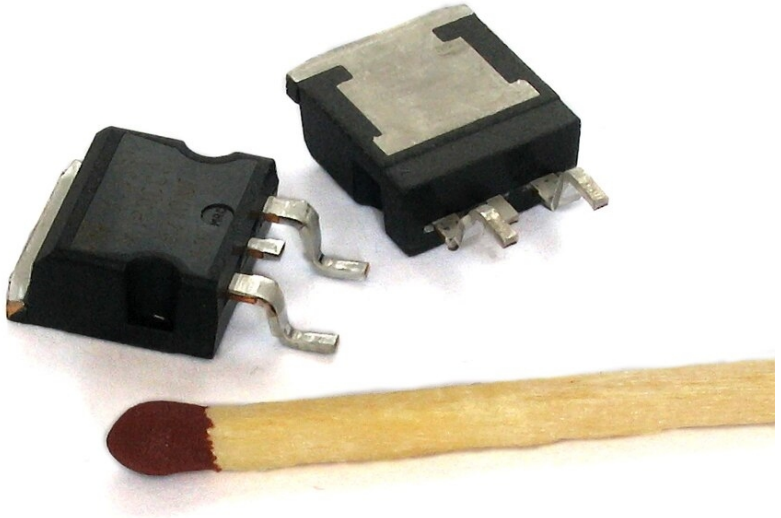
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Electronics & Precision Agriculture Laboratory | Academic Year 2025/2026 | Sokoine University of Agriculture

Lecture 6 Learning Outcomes & Practical Scope

- **Contrast Unipolar vs Bipolar Physics:** Voltage-controlled FET operation vs current-controlled BJT mechanics.
- **Analyze JFET Principles:** Understand pinch-off voltage V_P, saturation current I_{DSS}, and Shockley's equation.
- **Evaluate Transconductance g_m:** Derive $g_m = dI_D / dV_{GS}$ and determine operating point transconductance.
- **Analyze JFET Biasing Networks:** Design Self-Bias and Voltage Divider Bias configurations.
- **Differentiate MOSFET Topologies:** Depletion-Enhancement (DE-MOSFET) vs Enhancement-Only (E-MOSFET) devices.
- **Master Power MOSFET Dynamics:** Gate charge Q_G, threshold voltage $V_{GS(th)}$, ON-resistance $R_{DS(on)}$, and heat sinking.
- **Deploy Power MOSFETs in AgriTech:** Design solar brushless DC pump inverters and implement ESD protection protocols.

Field Effect Transistors in Modern Agro-Mechatronics

- **The Powerhouse of Solar Irrigation:** High-efficiency solar water pumping systems rely exclusively on Power MOSFETs.
- **Negligible Gate Drive Power:** Ultra-high input impedance (> 10^12 Ohm) allows microcontrollers to control kilowatts of motor power.
- **High-Speed PWM Actuation:** Power MOSFETs switch at 20 kHz to 100 kHz with minimal switching losses in variable-rate sprayers.
- **Superior Thermal Stability:** MOSFETs have a positive temperature coefficient of ON-resistance (R_DS(on)), preventing thermal runaway.
- **Harsh Agro-Climatic Vulnerability:** Insulated silicon dioxide (SiO2) gate dielectric is microscopically thin; easily punctured by static electricity.

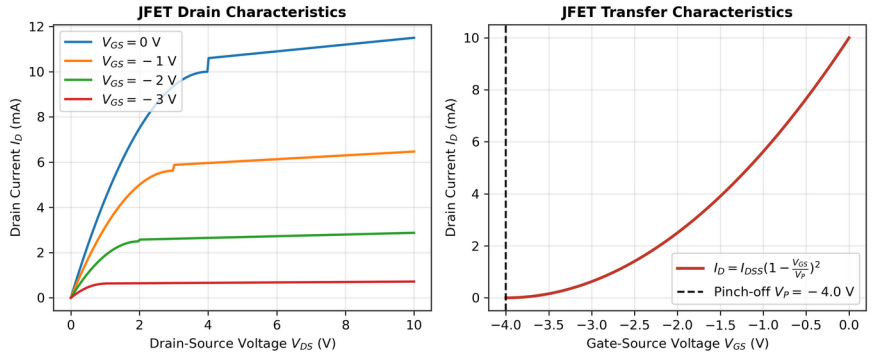


Fundamental Comparison: FET versus BJT

- **Control Mechanism:** BJT is current-controlled (I_C controlled by I_B); FET is voltage-controlled (I_D controlled by gate voltage V_GS).
- **Carrier Type:** BJT is bipolar (conduction involves both electrons and holes); FET is unipolar (conduction involves only majority carriers).
- **Input Impedance:** BJT input impedance is moderate (1 kOhm to 10 kOhm); FET input impedance is astronomical (10^9 to 10^14 Ohm).
- **Thermal Coefficient:** BJT has negative tempco (prone to thermal runaway); Power MOSFET has positive tempco (inherently self-stabilizing).
- **Switching Speed:** FETs have zero minority carrier storage time; switch orders of magnitude faster than power BJTs.
- **Noise Performance:** FETs exhibit substantially lower thermal and flicker noise; ideal for high-impedance agricultural sensor preamps.

Junction Field Effect Transistor (JFET) Physics

- **N-Channel JFET Construction:** Bar of N-type silicon with Drain (D) and Source (S) ohmic contacts at opposite ends.
- **Gate Electrodes:** Two heavily doped P^+ regions embedded on opposite sides of the channel, connected together as the Gate (G).
- **Normally-ON Depletion Device:** With V_GS = 0 V and small V_DS, majority electrons flow freely from source to drain through the open channel.
- **Reverse-Biased Gate Junction:** Applying negative gate voltage (V_GS < 0) widens the P^+N depletion regions into the channel, constricting channel width.
- **Voltage-Controlled Resistor (Ohmic Region):** At low V_DS, the channel behaves as a linear resistor whose resistance is modulated by V_GS.



Pinch-off Voltage V_P & Shockley's Transfer Equation

- **Pinch-Off Phenomenon:** As drain voltage V_DS increases, the drain-end depletion layers expand until they touch (pinch off) at V_DS = |V_P|.
- **Saturation (Active) Region:** Beyond pinch-off (V_DS >= V_GS - V_P), current saturates to a constant level governed strictly by gate voltage V_GS.
- **Pinch-Off Voltage (V_P):** Gate-to-source cutoff voltage that pinches channel completely shut at V_DS = 0 (typically -2V to -8V for N-channel).
- **Drain Saturation Current (I_DSS):** Maximum drain current flowing when V_GS = 0 V and V_DS >= |V_P|.
- **Shockley's Transfer Equation:** In saturation region: $I_D = I_{DSS} \cdot [\, 1 - (V_{GS} / V_P) \,]^2$.
- **Square-Law Characteristic:** Drain current varies quadratically with gate voltage; provides pure second-harmonic transfer without higher distortions.

JFET Small-Signal Transconductance g_m

- **Definition of Transconductance:** The rate of change of drain current with respect to gate-source voltage: g_m = dI_D / dV_GS |_(Q-point).
- **Mathematical Derivation:** Differentiating Shockley's equation: dI_D / dV_GS = 2 · I_DSS · [1 - (V_GS / V_P)] · (-1 / V_P).
- **Operating Transconductance Formula:** g_m = g_m0 · [1 - (V_GS / V_P)], where maximum transconductance g_m0 = (2 · I_DSS) / |V_P|.
- **Transconductance in Terms of Drain Current:** g_m = g_m0 · sqrt(I_D / I_DSS); Transconductance is proportional to square root of drain current.
- **Units:** Measured in Siemens (S), milliSiemens (mS), or micro-mhos (typically 1.0 mS to 10.0 mS for small-signal JFETs).

JFET Self-Bias Circuit Architecture

- **Single-Supply Biasing:** JFET requires negative V_{GS}, but Self-Bias generates this negative potential using a single positive supply V_{DD}.
- **Source Resistor Mechanism:** Source current I_D flows through resistor R_S, establishing a positive source voltage: $V_S = I_D \cdot R_S$.
- **Gate Resistor Grounding:** Large gate resistor R_G (1 MOhm to 10 MOhm) ties gate to ground: Since gate current $I_G \approx 0$, $V_G = 0$ V.
- **Resulting Negative Bias:** $V_{GS} = V_G - V_S = 0 - I_D \cdot R_S = - I_D \cdot R_S$.
- **Self-Regulating Action:** If I_D attempts to rise -> V_S rises -> V_{GS} becomes more negative -> constricts channel -> forces I_D back down!
- **Source Resistor Sizing:** $R_S = |V_{GSQ}| / I_{DQ}$; Sized from desired Q-point coordinates.

Worked Engineering Example 1: JFET Self-Bias Q-Point for Soil pH Probe Pre-Amp

PROBLEM SPECIFICATION & AGRICULTURAL CONTEXT:
A 2N5458 N-channel JFET buffers a high-impedance soil pH glass electrode sensor in a Morogoro agricultural chemistry lab. Manufacturer datasheet specifications: $I_{DSS} = 8.0$ mA and $V_P = -4.0$ V. The circuit uses self-bias with $V_{DD} = 18.0$ V, drain resistor $R_D = 2.2$ kOhm, and source resistor $R_S = 560$ Ohm. Gate resistor $R_G = 2.2$ MOhm. Determine: (a) Quiescent drain current I_{DQ}, (b) Quiescent gate-source voltage V_{GSQ}, (c) Quiescent drain-source voltage V_{DSQ}, (d) Transconductance g_m at Q-point.

STEP-BY-STEP MATHEMATICAL SOLUTION & VERIFICATION:

1. Formulate System of Equations: Equation 1 (Shockley): $I_D = I_{DSS} \cdot [\, 1 - (V_{GS} / V_P) \,]^2$; Equation 2 (Bias Line): $V_{GS} = - I_D \cdot R_S$.
2. Substitute V_{GS} into Shockley: $I_D = 8.0 \text{ mA} \cdot [\, 1 - (- I_D \cdot 560 / -4.0) \,]^2 = 0.008 \cdot [\, 1 - 140 \cdot I_D \,]^2$.
3. Expand Quadratic: $I_D = 0.008 \cdot [\, 1 - 280 \cdot I_D + 19{,}600 \cdot I_D^2 \,] = 0.008 - 2.24 \cdot I_D + 156.8 \cdot I_D^2$.
   Standard Quadratic: $156.8 \cdot I_D^2 - 3.24 \cdot I_D + 0.008 = 0$.
   Using quadratic formula: $I_{DQ} = [\, 3.24 - \text{sqrt}(3.24^2 - 4 \cdot 156.8 \cdot 0.008) \,] / (2 \cdot 156.8) = [\, 3.24 - 2.341 \,] / 313.6 = 0.002867$ A $= 2.87$ mA.
4. Calculate Voltages: $V_{GSQ} = - I_{DQ} \cdot R_S = - (2.87 \text{ mA} \cdot 560 \text{ Ohm}) = - 1.607$ V.
   $V_{DSQ} = V_{DD} - I_{DQ} \cdot (R_D + R_S) = 18.0 \text{ V} - 2.87 \text{ mA} \cdot (2{,}200 + 560) = 18.0 \text{ V} - 7.92 \text{ V} = 10.08$ V.
5. Calculate Operating Transconductance: $g_{m0} = (2 \cdot I_{DSS}) / |V_P| = (2 \cdot 8.0 \text{ mA}) / 4.0 \text{ V} = 4.0$ mS.
   $g_m = g_{m0} \cdot [\, 1 - (V_{GSQ} / V_P) \,] = 4.0 \text{ mS} \cdot [\, 1 - (-1.607 / -4.0) \,] = 4.0 \cdot (1 - 0.4018) = 2.39$ mS.

MOSFET Architecture: Insulated Gate Field Effect Physics

- **Insulated Gate Structure:** Metal gate electrode is electrically isolated from the semiconductor channel by an ultra-thin Silicon Dioxide (SiO2) layer.
- **Zero Gate Current:** Gate oxide dielectric resistance exceeds 10^14 Ohm; DC gate leakage current is measured in pico-Amperes (pA).
- **Depletion-Enhancement MOSFET (DE-MOSFET):** Built with a physical silicon channel; can operate in either depletion mode ($V_{GS} < 0$) or enhancement mode ($V_{GS} > 0$).
- **Enhancement-Only MOSFET (E-MOSFET):** No physical channel exists at zero bias; completely normally-OFF ($I_D = 0$ at $V_{GS} = 0$).
- **Channel Inversion:** Applying positive V_{GS} repels holes and attracts minority electrons to the surface, creating an inverted N-type conductive layer.
- **Threshold Voltage ($V_{GS(th)}$):** Minimum positive gate voltage required to form the conductive inversion channel (typically 1.5V to 4.0V).

Enhancement MOSFET Governing Equations

- **Cutoff Region ($V_{GS} < V_{GS(th)}$):** Inversion channel is not formed; drain current $I_D = 0$; switch is fully OPEN.
- **Ohmic (Triode / Linear) Region ($V_{GS} > V_{GS(th)}$ and $V_{DS} < V_{GS} - V_{GS(th)}$):** Channel conducts as a continuous resistor.
- **Ohmic Current Equation:** $I_D = k \cdot [\, 2 \cdot (V_{GS} - V_{GS(th)}) \cdot V_{DS} - V_{DS}^2 \,]$, where k is device transconductance parameter (A/V^2).
- **Saturation (Active) Region ($V_{DS} >= V_{GS} - V_{GS(th)}$):** Drain-end channel pinches off; current saturates into constant current source.
- **Saturation Current Equation:** $I_D = k \cdot [\, V_{GS} - V_{GS(th)} \,]^2$.
- **Device Constant k:** $k = 0.5 \cdot \mu_n \cdot C_{ox} \cdot (W / L)$, where C_{ox} is gate capacitance per unit area, and W/L is channel width-to-length ratio.

Power MOSFETs: Vertical Trench Architecture

- **Planar vs Vertical Conduction:** Small-signal FETs conduct horizontally across the chip surface; Power MOSFETs conduct vertically from top to bottom.
- **Vertical DMOS / Trench HEXFET:** Current flows vertically from source on the top surface down through epitaxial drift layer to drain on the bottom tab.
- **Massive Paralleled Cells:** Millions of microscopic hexagonal cells integrated in parallel on a single silicon die to achieve massive current ratings (50A to 200A+).
- **Ultra-Low ON-Resistance (R_DS(on)):** State-of-the-art power MOSFETs achieve R_DS(on) under 5 milli-Ohms (mOhm), minimizing conduction I^2R losses.
- **Integral Body Diode:** Inherent parasitic PN junction between P-body and N-drift layer forms an internal reverse flyback diode across Drain and Source.

Gate Charge Dynamics & Driver Circuit Requirements

- **The Gate Capacitance Myth:** MOSFET gates do not draw DC current, but they exhibit significant parasitic capacitance: C_iss = C_gs + C_gd.
- **Miller Capacitance (C_gd):** Gate-to-drain capacitance undergoes Miller multiplication during switching transitions, forming the **Miller Plateau**.
- **Total Gate Charge (Q_G):** Total electrical charge (nanocoulombs, nC) that must be transferred to charge gate to desired V_GS.
- **Switching Inrush Current:** To switch in t_sw = 50 nanoseconds, peak gate current required: I_gate(peak) = Q_G / t_sw (can exceed 2 to 5 Amperes!).
- **Dedicated Gate Driver ICs:** Microcontroller GPIO pins cannot supply 2A transients; dedicated gate driver ICs (TC4427, IR2110) are mandatory.

Logic-Level vs Standard Power MOSFET Selection

- **Standard Power MOSFETs (e.g. IRF540N):** Require full V_{GS} = 10.0 V to reach minimum rated ON-resistance $R_{DS(on)}$ (44 mOhm).
- **The 3.3V / 5V Gate Pitfall:** Driving a standard IRF540 directly from a 3.3V or 5V microcontroller pin leaves it in the high-resistance linear region!
- **Thermal Destruction:** At V_{GS} = 3.3V, R_{DS} surges to > 2 Ohm; a 10A solar pump motor causes $P = I^2R$ = 200W dissipation, incinerating the transistor.
- **Logic-Level Power MOSFETs (e.g. IRLZ44N, IRL540):** Engineered with ultra-thin gate oxides; guarantee minimum $R_{DS(on)}$ at V_{GS} = 4.5V or 3.3V.
- **Rule of Thumb:** For direct microcontroller GPIO driving, ALWAYS specify logic-level 'IRL' prefix rather than standard 'IRF' prefix devices.

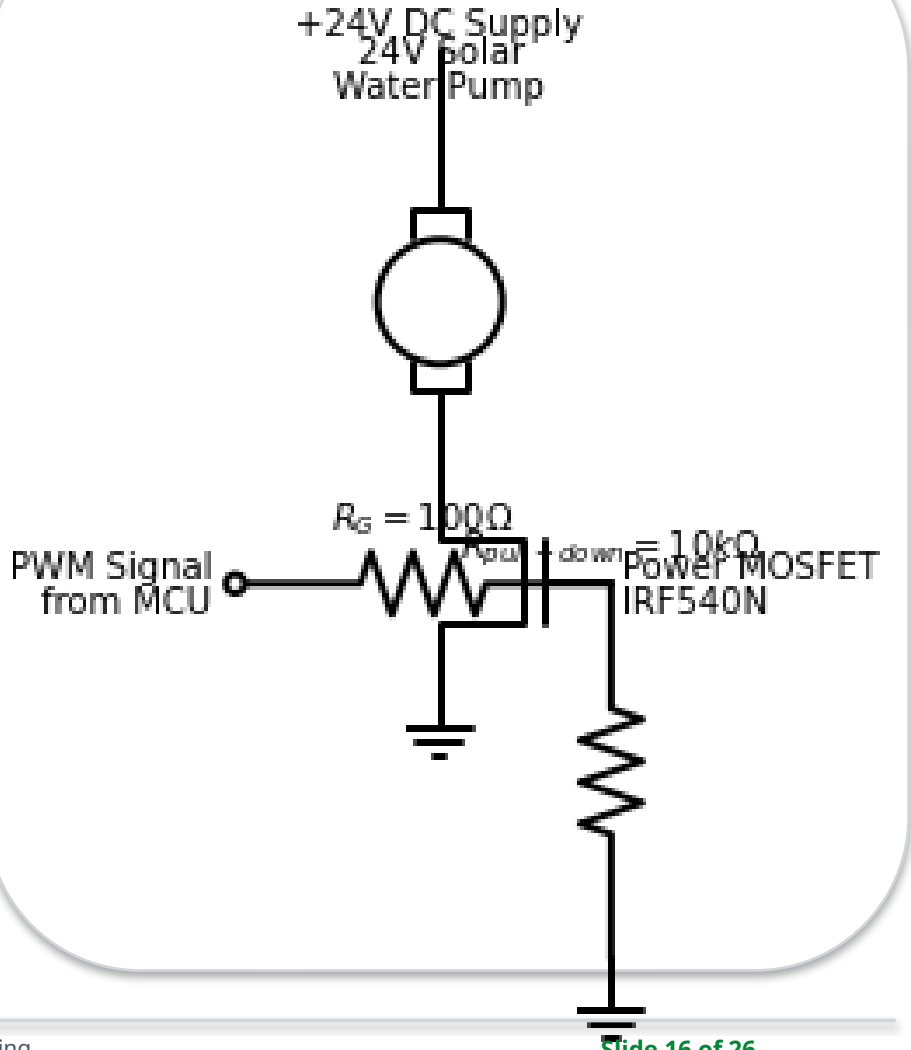
Worked Engineering Example 2: Power MOSFET Switch Sizing for Solar Water Pump

PROBLEM SPECIFICATION & AGRICULTURAL CONTEXT:
Design a low-side solid-state power switch for a 24.0 V, 120 W DC brushless solar irrigation pump at Bahi, Dodoma. Continuous motor current is I_load = 5.0 A. Ambient temperature reaches T_amb = 45°C. Compare: (a) Standard IRF540N driven at V_GS = 4.5 V (where R_DS(on) = 0.18 Ohm), versus (b) Logic-level IRLZ44N driven at V_GS = 4.5 V (where R_DS(on) = 0.028 Ohm). The TO-220 package has junction-to-ambient thermal resistance theta_JA = 62°C/W (free air without heatsink). Calculate power dissipations and junction temperatures.

STEP-BY-STEP MATHEMATICAL SOLUTION & VERIFICATION:
1. Standard IRF540N Analysis at V_GS = 4.5 V:
   Conduction Power Dissipation: $P_{IRF} = I_{load}^2 \cdot R_{DS(on)} = (5.0\ A)^2 \cdot 0.18\ Ohm = 25 \cdot 0.18 = 4.50\ W$.
   Junction Temperature Rise: $\Delta T_{IRF} = P_{IRF} \cdot \theta_{JA} = 4.50\ W \cdot 62°C/W = 279°C$.
   Total Junction Temperature: $T_{J(IRF)} = 45°C + 279°C = 324°C$ (catastrophic silicon die burnout! Max rating is 175°C).
2. Logic-Level IRLZ44N Analysis at V_GS = 4.5 V:
   Conduction Power Dissipation: $P_{IRL} = I_{load}^2 \cdot R_{DS(on)} = (5.0\ A)^2 \cdot 0.028\ Ohm = 25 \cdot 0.028 = 0.70\ W$.
   Junction Temperature Rise: $\Delta T_{IRL} = P_{IRL} \cdot \theta_{JA} = 0.70\ W \cdot 62°C/W = 43.4°C$.
   Total Junction Temperature: $T_{J(IRL)} = 45°C + 43.4°C = 88.4°C$ (completely safe, operates comfortably under free convection!).
3. Engineering Impact: Selecting the correct logic-level MOSFET eliminated heatsink cost and guaranteed reliable operation.

Power MOSFET H-Bridge & BLDC Motor Inverters

- **H-Bridge Architecture:** Four MOSFETs arranged in an H-network allow bidirectional forward, reverse, and regenerative braking of DC motors.
- **High-Side vs Low-Side Switches:** Low-side N-channel MOSFETs have grounded sources; high-side MOSFETs have sources floating at motor voltage.
- **High-Side Gate Bootstrapping:** High-side N-channel MOSFET requires gate voltage V_gate = V_bus + 10V; achieved via bootstrap capacitor and diode.
- **Three-Phase BLDC Inverter:** Six power MOSFETs configured into three half-bridge legs drive 3-phase permanent magnet solar submersible pumps.
- **Pulse Width Modulation (PWM):** Microcontroller switches MOSFETs at 20 kHz, modulating duty cycle to achieve variable-speed water pumping.



Electrostatic Discharge (ESD) Protection Protocols

- **The Dielectric Puncture Threat:** SiO2 gate dielectric is only 5 to 50 nanometers thick; punctures permanently at V_GS > 20 V to 30 V.
- **Human Body Model (HBM):** A technician walking across a nylon carpet or dry African grain floor accumulates static potentials exceeding 15,000 V!
- **Invisible Latent Damage:** Minor ESD zap creates microscopic gate oxide pinholes; device passes bench test but fails prematurely in the field.
- **Internal Protection Diodes:** Many modern small-signal MOSFETs incorporate back-to-back Zener diodes across Gate and Source to clamp ESD.
- **Field Handling Protocol:** Wear grounded antistatic wrist straps; store MOSFETs in conductive black ESD foam; solder with grounded-tip irons.

Paralleling Power MOSFETs: The Positive Tempco Advantage

- **The BJT Current Hogging Dilemma:** BJTs have negative tempco (hotter transistor draws more current), causing thermal runaway when paralleled.
- **MOSFET Self-Balancing Mechanism:** Silicon mobility decreases with temperature, causing R_DS(on) to increase as temperature rises (+0.6%/°C).
- **Current Sharing Dynamics:** If MOSFET A runs hotter than MOSFET B -> R_DS(on) of A increases -> current automatically diverts into B!
- **Direct Paralleling:** Multiple power MOSFETs can be connected in parallel without external emitter ballasting resistors to handle hundreds of Amps.
- **Individual Gate Resistors:** Always provide a dedicated 10 Ohm to 47 Ohm series gate damping resistor for each parallel MOSFET to prevent parasitic HF ringing.

MOSFET Circuit Hardening for Rural Agriculture

- **Gate Pulldown Resistors:** Always place a 10 kOhm to 100 kOhm resistor directly between Gate and Source to ensure OFF state during MCU boot.
- **Flyback Snubbers:** Pair integral body diodes with external RC snubbers (10 Ohm + 4.7 nF) to suppress high-frequency inductive motor ringing.
- **Overvoltage Clamping:** Solder a bidirectional 18V Zener or TVS diode directly across Gate-Source pins on the PCB.
- **Thermal Heatsinking:** For motor currents exceeding 10A, mount MOSFETs to finned aluminum heatsinks with thermal silicone insulator pads.
- **Tropical Humidity Ingress:** Encapsulate high-voltage gate drive traces in conformal polyurethane coating to prevent ionic surface leakage.

Tanzanian Field Case Study 1: Solar Submersible Pump MOSFET Inverter Failure

FIELD LOCATION & PROBLEM IDENTIFICATION

• **Geographic Location: Mbarali Paddy Rice Irrigation Scheme, Mbeya Region, Tanzania**

A 7.5 kW solar photovoltaic submersible borehole pumping station suffered complete inverter blowouts every 4 to 6 weeks. Three half-bridge power MOSFET modules (IRFP260N) repeatedly suffered explosive drain-to-source punch-through, cutting off irrigation water to 40 hectares of rice paddies.

SUA ENGINEERING SOLUTION & AGRONOMIC IMPACT

SUA mechatronics engineers conducted DSO trace diagnostics and discovered two fatal design defects: (1) Inverter gate driver used standard 5V logic pulses that under-drove the 10V gates, causing excessive conduction heating, and (2) Long 80-meter submerged motor cables generated massive inductive voltage reflections (> 280V) that exceeded the 200V MOSFET breakdown rating.

Measurable Performance Metrics:

✔️ Retrofitted with 500V, 0.075 Ohm CoolMOS power transistors with 12V high-current gate drivers.

✔️ Installed dv/dt output line choke filters to suppress motor cable voltage reflections.

✔️ Inverter operated 100% reliably throughout two entire 120-day paddy rice seasons.

Tanzanian Field Case Study 2: Precision PWM Variable-Rate Fertilizer Injection

FIELD LOCATION & PROBLEM IDENTIFICATION

• Geographic Location: Kilombero Valley Teak & Sugar Plantations, Morogoro, Tanzania

Tractor-mounted liquid fertilizer injection systems required high-speed variable-rate solenoid valve pulsing (100 Hz PWM) linked to tractor GPS forward velocity. The original mechanical relay switches failed within 3 days due to contact arcing, causing uneven fertilizer distribution and crop lodging.

SUA ENGINEERING SOLUTION & AGRONOMIC IMPACT

Engineers designed a solid-state PWM driver module using logic-level IRLB8721 power MOSFETs (R_DS(on) = 8.7 mOhm) controlled directly by an STM32 microcontroller. A fast ultra-fast recovery diode (MUR420) was placed across the 12V, 4A fertilizer solenoid valve to handle the 100 Hz inductive freewheeling current.

Measurable Performance Metrics:

✔️ Achieved over 25 million valve switching cycles without a single electrical failure.

✔️ Fertilizer application accuracy improved to within ±1.5% across varying tractor speeds.

✔️ Reduced liquid fertilizer wastage by 18%, saving the estate over 22,000 USD annually.

Laboratory Bench Protocol: JFET Transfer Curves & Power MOSFET Switching Bench Protocol

PRACTICAL LAB LEARNING OUTCOMES & BENCH REQUIREMENTS:

- Measure N-channel JFET (2N5458) transfer characteristics and extract I_DSS and pinch-off V_P.
- Characterize JFET self-bias circuit and verify operating Q-point against theoretical equations.
- Demonstrate gate threshold voltage V_GS(th) and switching action of logic-level power MOSFET.
- Drive a DC motor with PWM signal and observe V_DS switching waveforms and body diode clamping on DSO.

STEP-BY-STEP PROTOTYPING & DIAGNOSTIC PROCEDURE:

Step 1: Wire 2N5458 JFET on breadboard; connect variable DC supply across Drain-Source (V_DS = 10V).

Step 2: Sweep gate voltage V_GS from 0V to -5V using second DC supply; record I_D; plot curve and extract I_DSS and V_P.

Step 3: Assemble self-bias circuit (R_D = 2.2k, R_S = 470, R_G = 1M); measure V_GSQ and V_DSQ with DMM; verify math.

Step 4: Mount IRLZ44N power MOSFET; connect 12V DC cooling fan as load; apply 0V then 3.3V to gate; record V_DS.

Step 5: Inject 10 kHz PWM signal from function generator into gate; connect DSO across drain; observe switching rise/fall times.

Diagnostic Troubleshooting of FET Circuits

- **Floating Gate Latch-Up:** A disconnected MOSFET gate acts as an antenna, floating to random voltages and oscillating; always tie with pulldown.
- **Insufficient Gate Drive Voltage:** Driving standard MOSFETs with 3.3V logic causes extreme conduction heating and thermal failure.
- **Punctured Gate Dielectric:** DMM resistance between Gate and Source reads < 100 kOhm (should be infinite); caused by ESD or inductive spike.
- **Body Diode Reverse Recovery Ringing:** Slow internal body diode in bridge inverters creates shoot-through currents; add external Schottky bypass.
- **Parasitic Gate Oscillation:** High-speed switching induces high-frequency VHF ringing on gate lead; cure by adding 22 Ohm series gate resistor.

Lecture 6 Self-Assessment & Upcoming Topics

SELF-ASSESSMENT & PROBLEM-SOLVING EXERCISES:

Q: Differentiate between a JFET and a MOSFET in terms of physical structure, input impedance, and gate conduction mechanism.

Q: State Shockley's equation for a JFET in saturation and explain the physical significance of parameters I_DSS and V_P.

Q: Explain why a power MOSFET has a positive temperature coefficient of ON-resistance, and why this allows safe direct paralleling.

Q: Why is it hazardous to drive a standard power MOSFET (e.g. IRF540N) directly from a 3.3V microcontroller GPIO pin?

Q: Explain the Miller effect in power MOSFET switching and describe why the gate voltage plateaus during turn-on.

NEXT LECTURE PREVIEW & REQUIRED PREPARATION:

Lecture 7 will explore Number Systems & Binary Arithmetic: Radix conversions, signed 2's complement, BCD, and tractor CAN-bus codes.

Lecture 6 Synthesis: Key Takeaways & Formula Sheet

CORE CONCEPTUAL TAKEAWAYS

✔️ FETs are unipolar voltage-controlled devices with astronomical input impedance (> 10^12 Ohm).

✔️ JFETs are normally-ON depletion devices; channel conduction is governed by Shockley's square-law equation.

✔️ Power MOSFETs (HEXFETs) feature vertical trench structures with ultra-low ON-resistance ($R_{DS(on)} < 10$ mOhm).

✔️ Logic-level MOSFETs (IRLZ series) are mandatory for direct 3.3V / 5V microcontroller switching.

✔️ Thin SiO2 gate dielectrics are highly susceptible to ESD damage; strict static handling and gate TVS clamping are required.

CRITICAL ENGINEERING FORMULAS

★ JFET Shockley Equation: $I_D = I_{DSS} \cdot [\ 1 - (V_{GS} / V_P)\ ]^2$

★ Maximum Transconductance: $g_{m0} = (2 \cdot I_{DSS}) / |V_P|$

★ Operating Transconductance: $g_m = g_{m0} \cdot [\ 1 - (V_{GS} / V_P)\ ]$

★ JFET Self-Bias: $V_{GS} = - I_D \cdot R_S$

★ E-MOSFET Saturation: $I_D = k \cdot [\ V_{GS} - V_{GS(th)}\ ]^2$

★ MOSFET Conduction Loss: $P_{cond} = I_{D(rms)}^2 \cdot R_{DS(on)}$

★ Peak Gate Current: $I_{gate(peak)} = Q_G / t_{sw}$

Prescribed References & Practical Laboratory Preparation

- Boylestad, R. L., & Nashelsky, L. (2018). Electronic Devices and Circuit Theory (11th ed.). Pearson Education.
- Sedra, A. S., & Smith, K. C. (2020). Microelectronic Circuits (8th ed.). Oxford University Press.
- **Rashid, M. H. (2014). Power Electronics:** Devices, Circuits, and Applications (4th ed.). Pearson.

- **Laboratory Preparation:** Review antistatic ESD handling procedures before attending Lab Session 6.
- **Field Assignment:** Complete Assignment 2, Questions 9 to 12 on JFET biasing and power MOSFET inverter design.